Glossary

Term	Definition (Source: docs.docker.com/glossary)
container	A container is a runtime instance of a docker image. Containers are isolated processes, not virtual machines. A Docker container consists of a Docker image, an execution environment and, a standard set of instructions.
cgroups	cgroups is a Linux kernel feature that limits, accounts for, and isolates the resource usage (CPU, memory, disk I/O, network, etc.) of a collection of processes. Docker relies on cgroups to control and isolate resource limits.
Docker	The Docker project as a whole, which is a platform for developers and sysadmins to develop, ship, and run apps. Docker is an open-source utility that automates the deployment and management of programs inside software containers. The docker daemon process running on the host which manages images and containers (also called Docker Engine).
Docker Compose	Compose is a tool for defining and running complex applications with Docker.
Docker Hub	The Docker Hub is a centralized resource for working with Docker and its components.
Dockerfile	A Dockerfile is a text document that contains all the commands you would normally execute manually in order to build a Docker image. Docker can build images automatically by reading the instructions from a Dockerfile.
Docker Machine	Docker tool which makes it really easy to create Docker hosts on your computer, on cloud providers and inside your own DC. It creates servers, installs Docker on them, then configures the Docker client to talk to them.
Docker Swarm	Docker Swarm is the name of a standalone native clustering tool for Docker. Docker Swarm pools together several Docker hosts and exposes them as a single virtual Docker host. It serves the standard Docker API, so any tool that already works with Docker can now transparently scale up to multiple hosts.
image	Docker images are the basis of containers. An Image is an ordered collection of root filesystem changes and the corresponding execution parameters for use within a container runtime. An image typically contains a union of layered filesystems stacked on top of each other. An image does not have state and it never changes.
namespace	namespace is a Linux kernel feature that isolates and vitualizes system resources. Namespaces are an important part of Docker's isolation model.
node	A node is a physical or virtual machine running an instance of the Docker Engine in swarm mode.
registry	A Registry is a hosted service containing repositories of images which responds to the Registry API. The default registry can be accessed using a browser at Docker Hub or using the docker search command.
repository	A repository is a set of Docker images. A repository can be shared by pushing it to a registry server. The different images in the repository can be labeled using tags.
swarm	A swarm is a cluster of one or more Docker Engines running in swarm mode.
Swarm mode	Swarm mode refers to cluster management and orchestration features embedded in Docker Engine. When you initialize a new swarm (cluster) or join nodes to a swarm, the Docker Engine runs in swarm mode.
Union file system (UnionFS)	UnionFS implement a union mount and operate by creating layers. Docker uses UnionFS+copy-on-write to provide the building blocks for containers, making them lightweight & fast.

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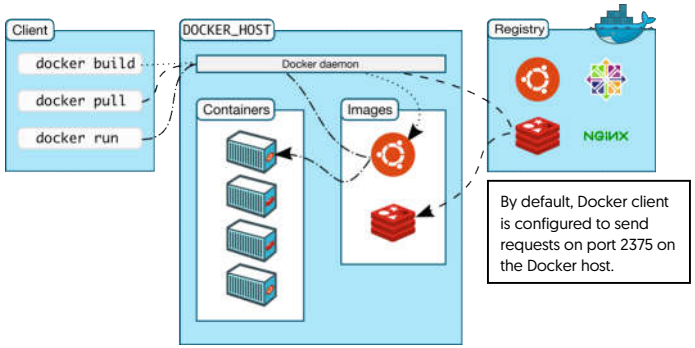
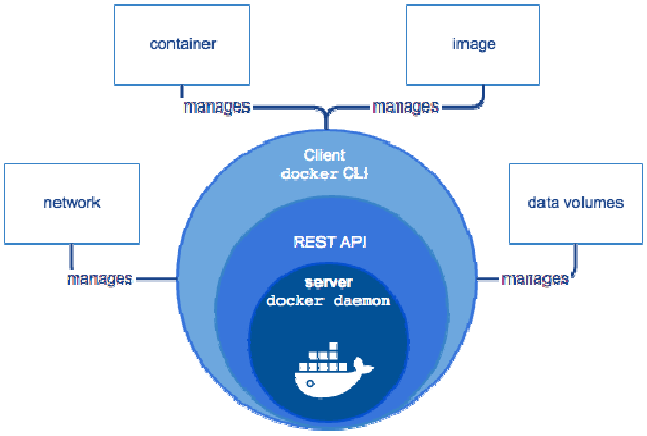
Docker Architecture

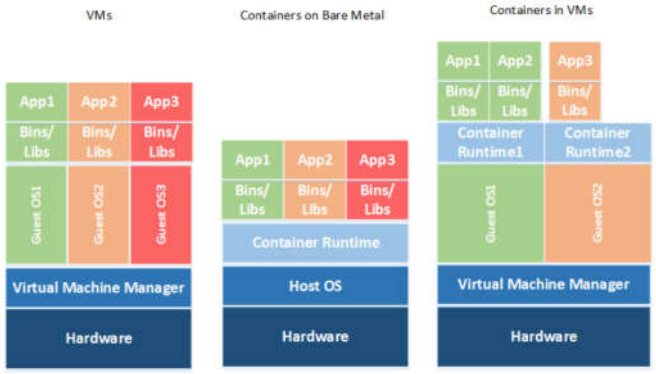



```
$ docker –H <IP>:<Port> <Command>
$ export DOCKER_HOST=IP:Port
$ docker <Command>
```

Docker Engine



Virtual Machine vs. Container Arch.



docker

Cheat Sheet v.1.0

```
# curl –fsSL https://get.docker.com/ | sh
# docker version
```

vjavierf.wordpress.com

Updated May 2017

Docker Management Commands

Command	Description
docker container	Manage containers
docker image	Manage images
docker network	Manage networks
docker node	Manage Swarm nodes
docker plugin	Manage plugins
docker secret	Manage Docker secrets
docker service	Manage services
docker stack	Manage Docker stacks
docker swarm	Manage Swarm
docker system	Manage Docker
docker volume	Manage volumes

docker container Commands

Command	Description
attach	Attach to a running container
commit	Create a new image from a container's changes
cp	Copy files/folders between a container and the local filesystem
create	Create a new container
diff	Inspect changes to files or directories on a container's filesystem
exec	Run a command in a running container
export	Export a container's filesystem as a tar archive
inspect	Display detailed information on one or more containers
kill	Kill one or more running containers
logs	Fetch the logs of a container
ls	List containers
pause	Pause all processes within one or more containers
port	List port mappings or a specific mapping for the container
prune	Remove all stopped containers
rename	Rename a container
restart	Restart one or more containers
rm	Remove one or more containers
run	Run a command in a new container
start	Start one or more stopped containers
stats	Display a live stream of container(s) resource usage statistics
stop	Stop one or more running containers
top	Display the running processes of a container
unpause	Unpause all processes within one or more containers
update	Update configuration of one or more containers
wait	Block until one or more containers stop, then print their exit codes

docker image Commands

Command	Description
build	Build an image from a Dockerfile
history	Show the history of an image
import	Import the contents from a tarball to create a filesystem image
inspect	Display detailed information on one or more images
load	Load an image from a tar archive or STDIN
ls	List images
prune	Remove unused images
pull	Pull an image or a repository from a registry
push	Push an image or a repository to a registry
rm	Remove one or more images
save	Save one or more images to a tar archive (streamed to STDOUT by default)
tag	Create a tag TARGET_IMAGE that refers to SOURCE_IMAGE

docker network Commands

Command	Description
connect	Connect a container to a network
create	Create a network
disconnect	Disconnect a container from a network
inspect	Display detailed information on one or more networks
ls	List networks
prune	Remove all unused networks
rm	Remove one or more networks

docker node Commands

Command	Description
demote	Demote one or more nodes from manager in the swarm
inspect	Display detailed information on one or more nodes
ls	List nodes in the swarm
promote	Promote one or more nodes to manager in the swarm
ps	List tasks running on one or more nodes, defaults to current node
rm	Remove one or more nodes from the swarm
update	Update a node

docker plugin Commands

Command	Description
create	Create a plugin from a rootfs and configuration
disable	Disable a plugin
enable	Enable a plugin
inspect	Display detailed information on one or more plugins
install	Install a plugin
ls	List plugins
push	Push a plugin to a registry

docker plugin Commands

Command	Usage
Rm	Remove one or more plugins
set	Change settings for a plugin
Upgrade	Upgrade an existing plugin

docker secret Commands

Command	Description
create	Create a secret from a file or STDIN as content
inspect	Display detailed information on one or more secrets
ls	List secrets
rm	Remove one or more secrets

docker stack Commands

Command	Description
deploy	Deploy a new stack or update an existing stack
ls	List stacks
ps	List the tasks in the stack
rm	Remove the stack
services	List the services in the stack

docker swarm Commands

Command	Description
init	Initialize a swarm
join	Join a swarm as a node and/or manager
join-token	Manage join tokens
leave	Leave the swarm
unlock	Unlock swarm
unlock-key	Manage the unlock key
update	Update the swarm

docker system Commands

Command	Description
df	Show docker disk usage
events	Get real time events from the server
info	Display system-wide information
prune	Remove unused data

docker volume Commands

Command	Description
create	Create a volume
inspect	Display detailed information on one or more volumes
ls	List volumes
Prune	Remove all unused volumes
rm	Remove one or more volumes